

Applied ML SP4 2025

Data collection and annotation, Imbalanced data, and Hyperparameters tuning

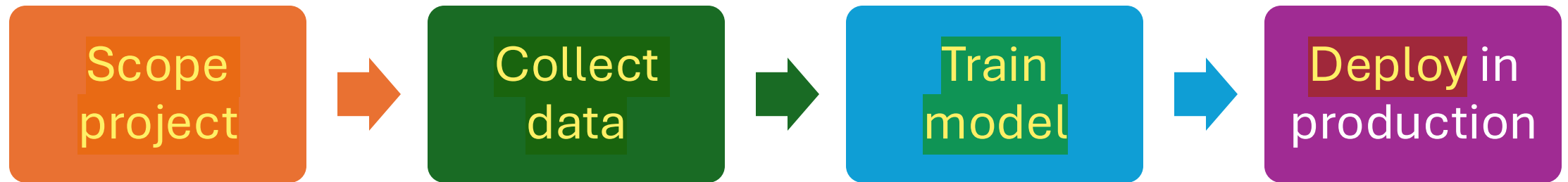
Selpi

Computer Science and Engineering
Chalmers and GU

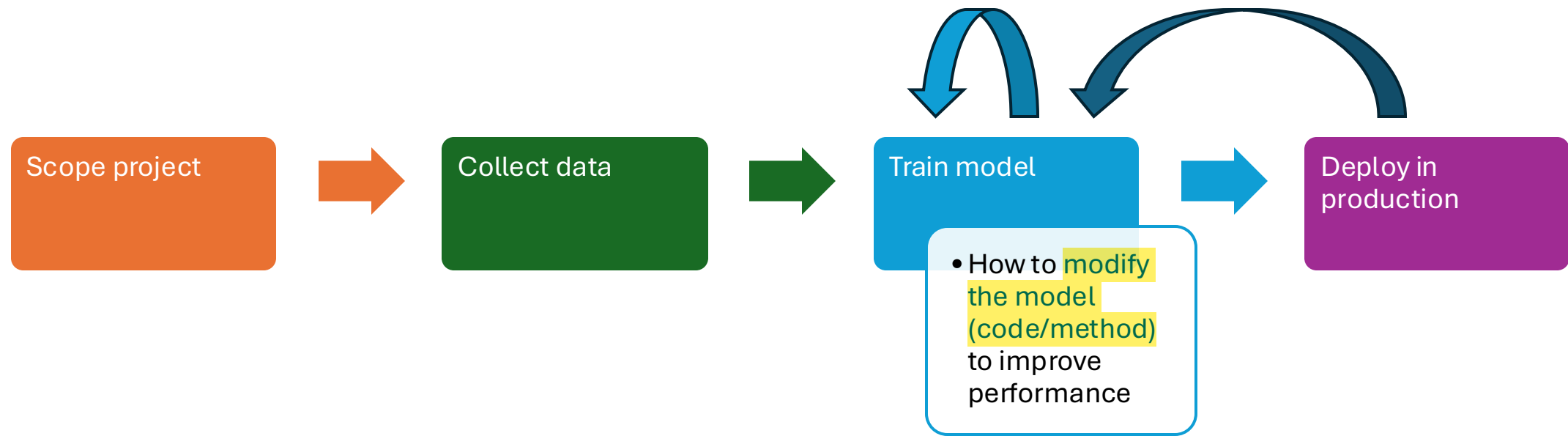
Data-centric *AI*

Simplified machine learning (ML) pipeline

AI System = Data + Code (Model)



Model-centric AI



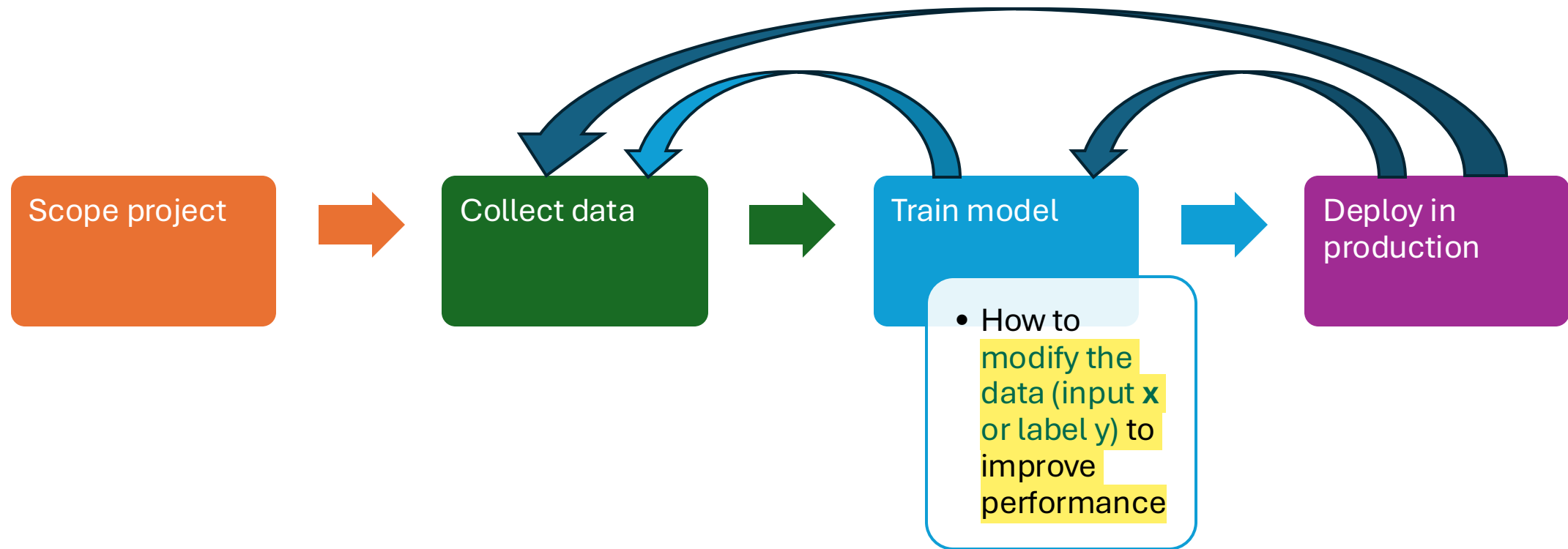
Data-centric AI

- A movement started by Andrew Ng in March 2021 ([link](#))
- It is a discipline of systematically engineering the data used to build an AI system
- Motivations:
 - Performance of ML systems depend on quality of data
 - Shift focus on data rather than only on models
 - Methods for model building are pretty established now
 - Abundance of data does not guarantee the quality.

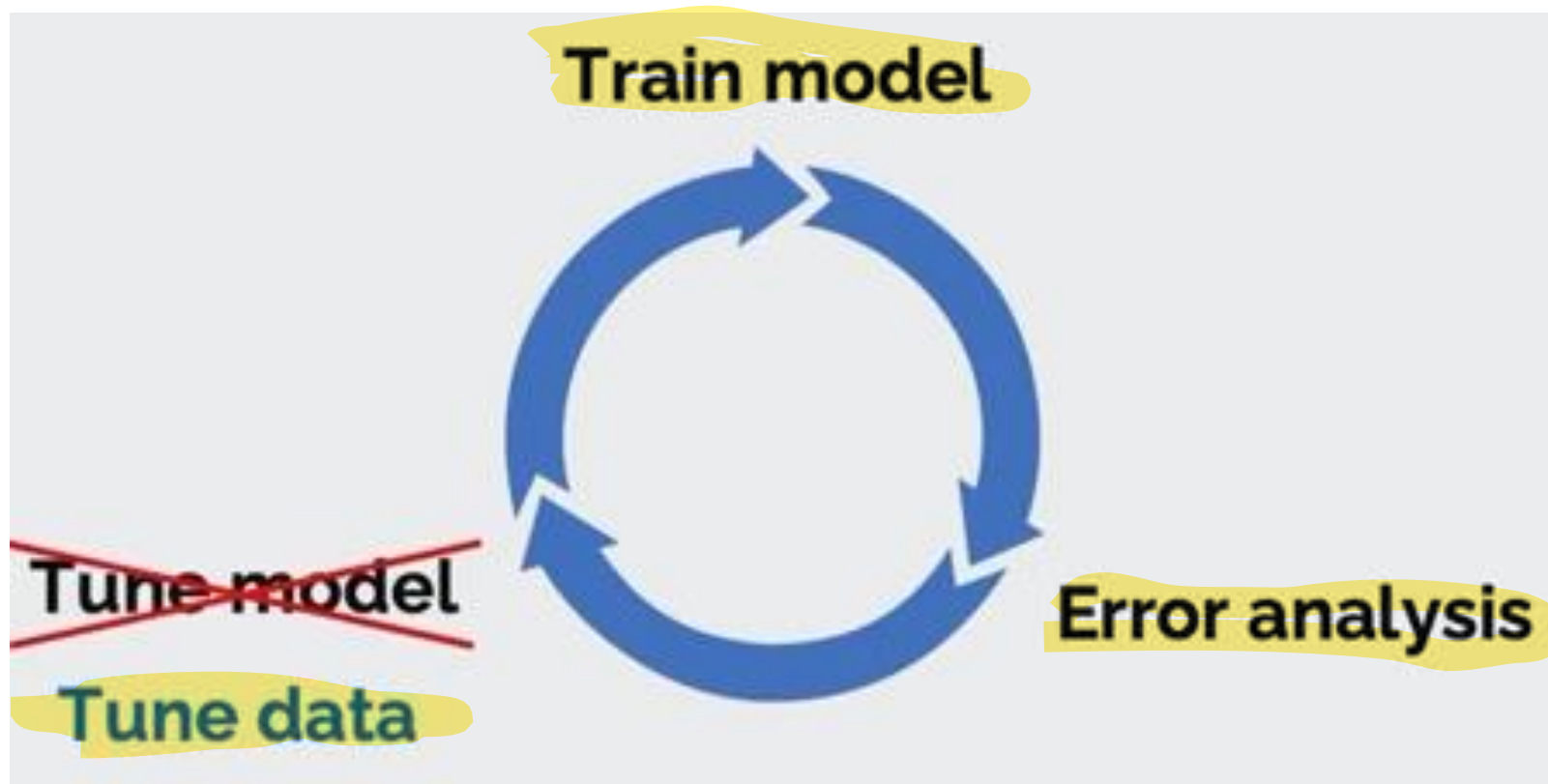
AI System = Data + Code (Model)

- Real cases suggest the need for systematic methods to clean, annotate, synthesise, and evaluate the data used to train and test models

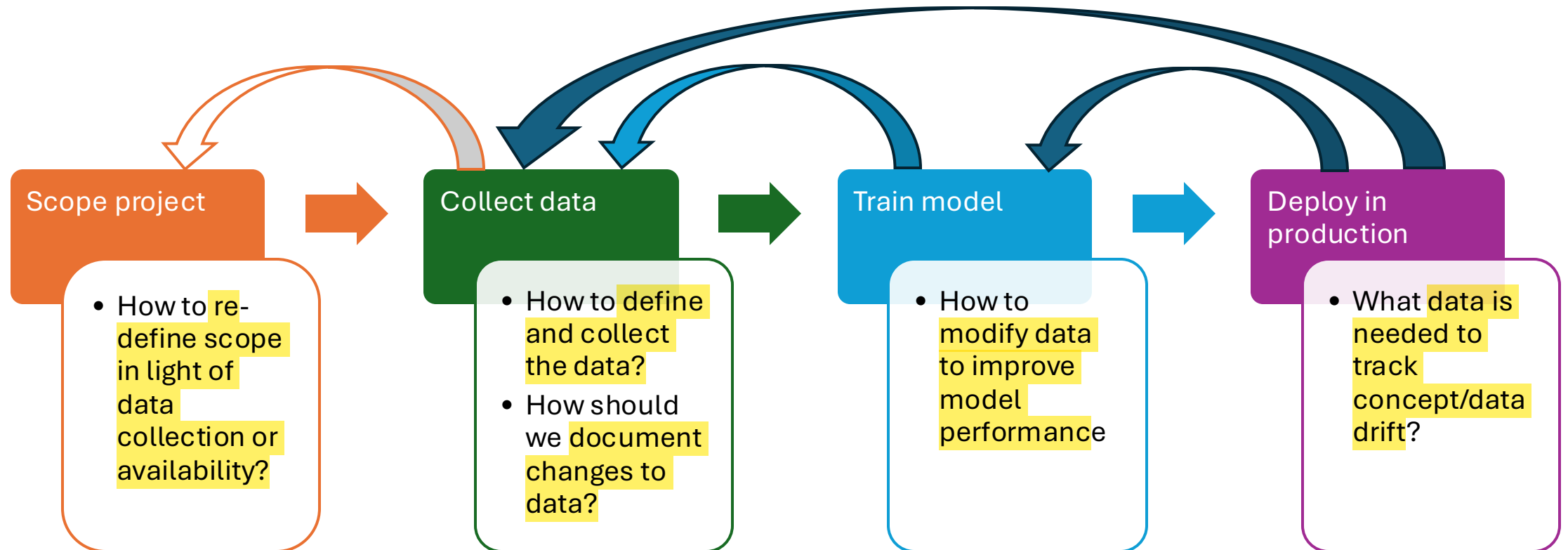
Data-centric AI



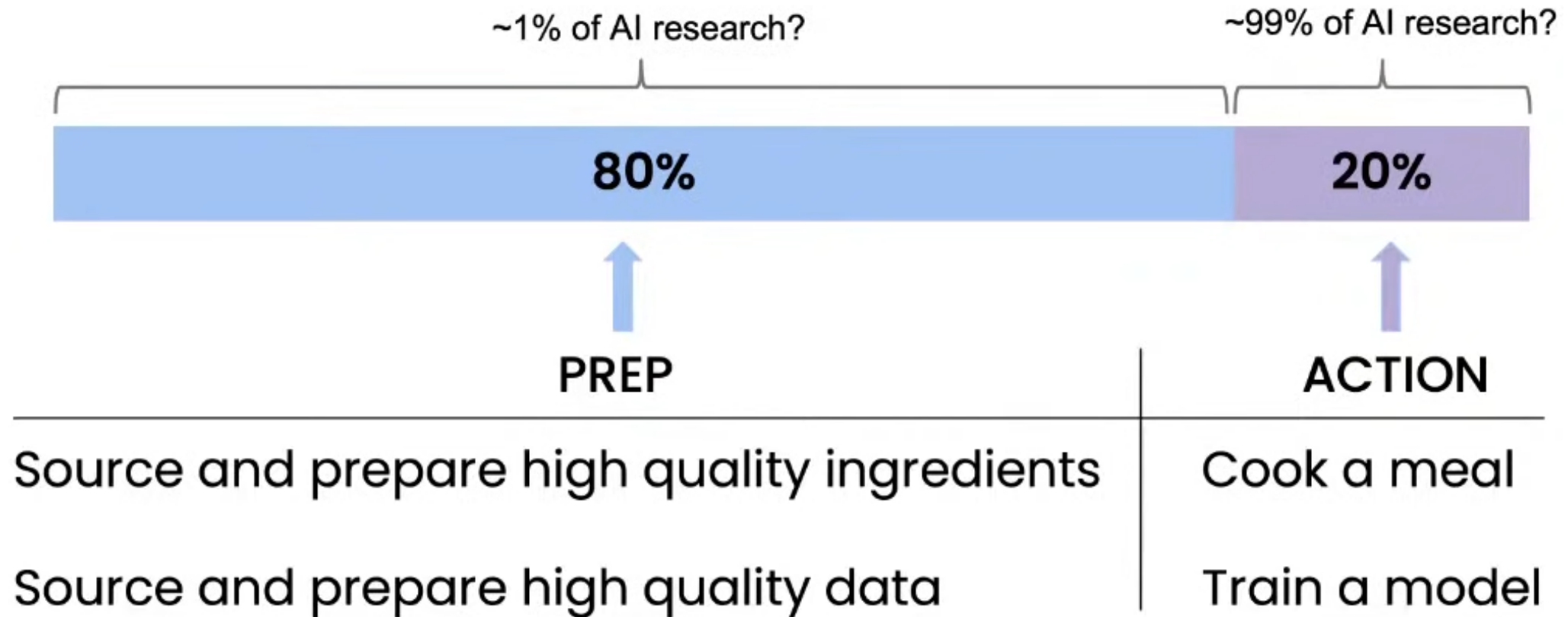
Data engineering as part of ML workflow



Data-centric AI → Good data in all phases of ML project lifecycle



Shift focus on data rather than only on model: Data is food for AI



Improve the code vs the data

	Steel defect detection	Solar panel	Surface inspection
Baseline	76.2%	75.68%	85.05%
Model-centric	+0% (76.2%)	+0.04% (75.72%)	+0.00% (85.05%)
Data-centric	+16.9% (93.1%)	+3.06% (78.74%)	+0.4% (85.45%)

Data collection and annotation

A case study to discuss

A group of researchers want to understand how drivers use their eyes while driving on real roads. They decided to equip 50 cars with sensors and data acquisition system to automatically collect driving data including video of drivers in the cabin whenever the car is on. To get more driving data, they plan the data collection to be done for 6 weeks in summer vacation period July-mid August this year.

To get the cars and drivers, they work together with a luxury car company. The company helps them to find cars and drivers from among its customer database. In the end they get 48 men and 2 women who are car enthusiasts joining the project.

You are in the advisory board of the project. Your task is to give advice on the study design made so far. What comments would you give to the researchers in the project.

Q: Are there any potential issues that should be pointed out, with respect to the purpose of the study? If there are, suggest how to improve this study design. If there aren't, give justification.

Aspects to consider for data collection

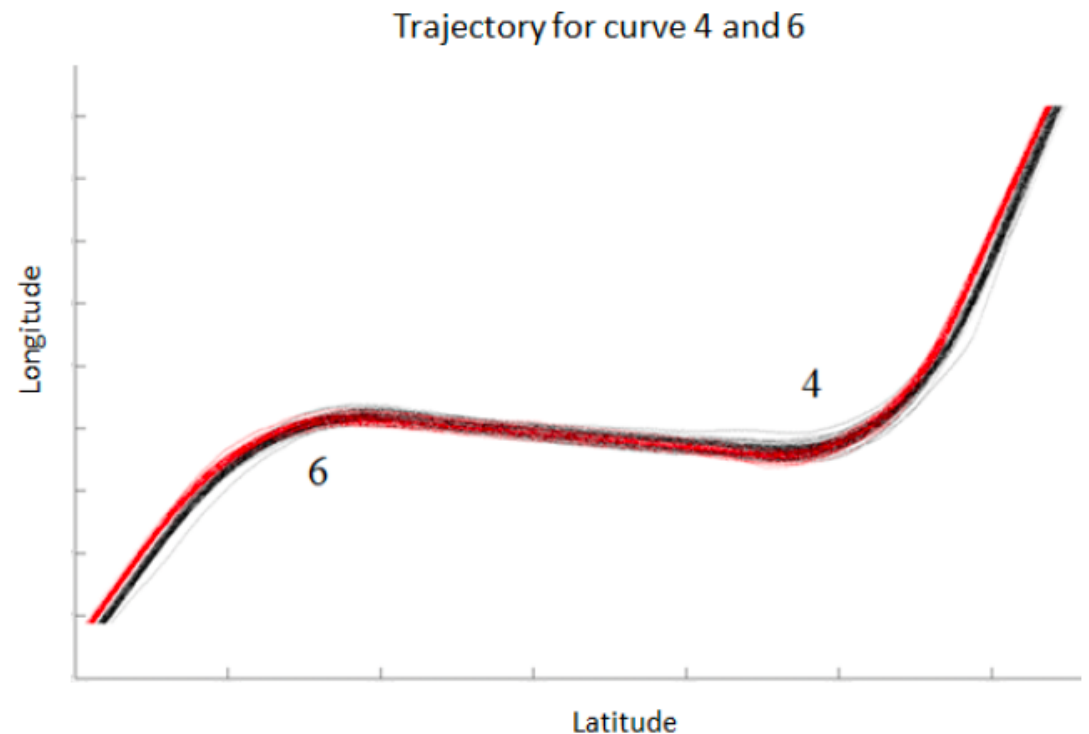
- Purpose of data collection
- Assumption
- Representativeness
- Sampling bias, non-response bias
- Stratification, weighting
- Availabilities (budget, time)
- Copyright issues, Privacy issues, Ethical issues, Legal issues
- Sampling effect on ML systems (time, selection, demography)

Data collection

- Data request is not approved
- Data governance and compliance e.g., Policy that location data cannot be taken outside the country
- Ethics approval takes longer than expected
- Experimental design had to be changed
- Difficult to recruit participants
- Placement of sensor affects quality of data

Data collection

- Measurement errors/quality of sensors
- Volume:
 - Data collected becomes much smaller than expected
 - Even if the data size is big, it may only have limited events of interest
- Not everything go according to plan – however we planned it.

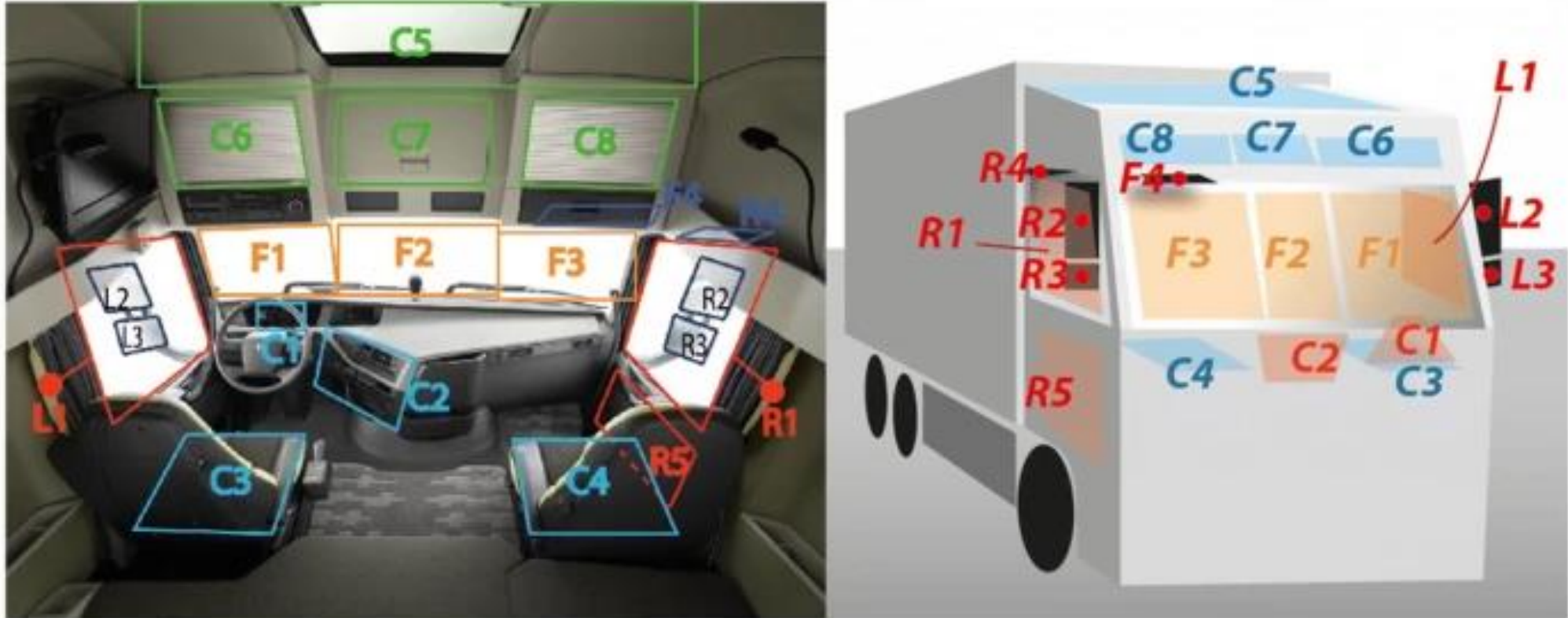


Annotation

Continuing from the case study:

- Assume now we have collected the data
- We now want to annotate where driver look (on road, off road, on the different locations such as left mirror, right mirror, center stack, etc...) – like in the paper “[Does agreement mean accuracy?....](#)”.
- Imagine for a few minutes **how** such an annotation might be carried out.
- What might be the **issues** to consider here?

Agreement does not mean accuracy!



Twenty glance locations from a cabin perspective (*left panel*) and an outside perspective (*right panel*).

Prefixes: L = Left, R = Right, F = Front, C = Cabin

PROS AND CONS OF LABELING APPROACHES

Approach	Description	Pros	Cons
Internal labeling	Assignment of tasks to an in-house data science team	✓ Predictable results ✓ High accuracy of labeled data ✓ The ability to track progress	✗ It takes much time
Outsourcing	Recruitment of temporary employees on freelance platforms, posting vacancies on social media and job search sites	✓ The ability to evaluate applicants' skills	✗ The need to organize workflow
Crowdsourcing	Cooperation with freelancers from crowdsourcing platforms	✓ Cost savings ✓ Fast results	✗ Quality of work can suffer
Specialized outsourcing companies	Hiring an external team for a specific project	✓ Assured quality	✗ Higher price compared to crowdsourcing
Synthetic labeling	Generating data with the same attributes of real data	✓ Fewer constraints for using sensitive and regulated data ✓ Training data without mismatches and gaps ✓ Cost- and time-effectiveness	✗ High computational power required
Data programming	Using scripts that programmatically label data to avoid manual work	✓ Automation ✓ Fast results	✗ Lower quality dataset

Annotation

- How can annotation influence the performance of ML systems?

Annotation quality affects the quality of training data and thus the performance of ML systems.

Wrong labels in benchmark datasets



ImageNet given label:
tick

Cleanlab guessed: **yellow garden spider**



ImageNet given label:
tick

Cleanlab guessed: **scorpion**



ImageNet given label:
red panda

Cleanlab guessed: **meerkat**



ImageNet given label:
red panda

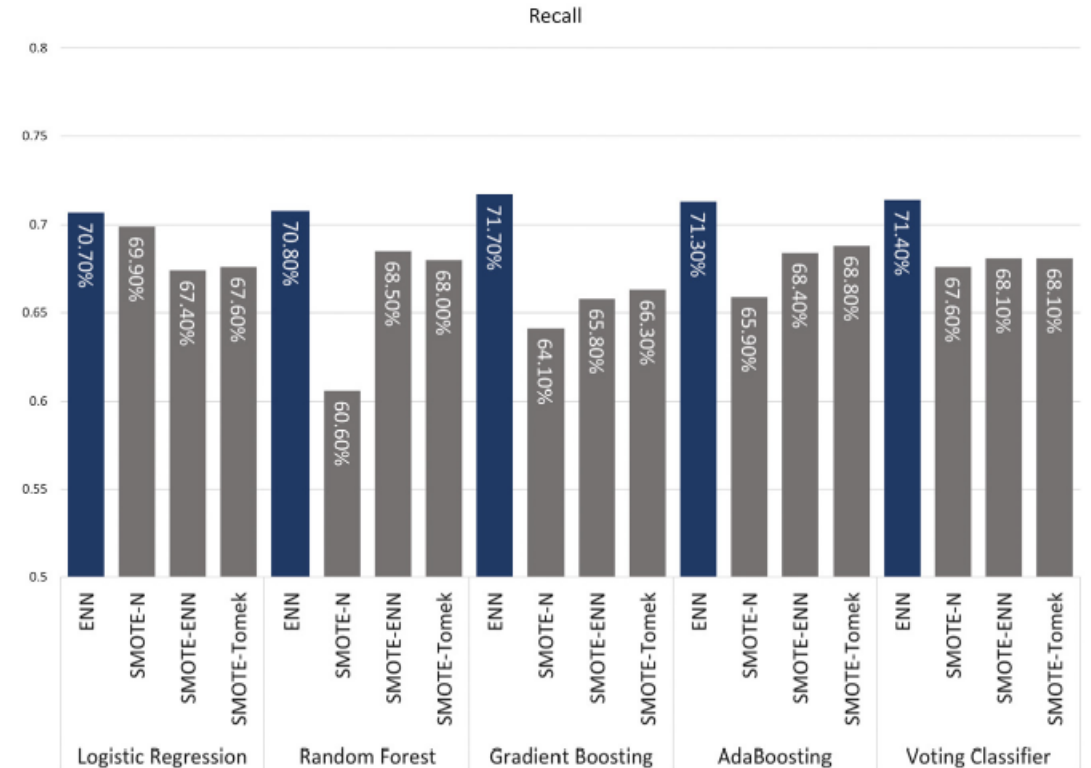
Cleanlab guessed: **giant panda**

Imbalanced data

Dealing with imbalanced data

Imbalanced data

- Diabetes data (type1, type 2)
- # Samples: almost 263 k
- Imbalanced: 18% positives
- Baseline recall: 57-58%
- ENN: undersampling majority class
- SMOTE-N: oversampling minority class
- SMOTE-ENN & SMOTE-Tomek : hybrid



Missing data, adding domain knowledge

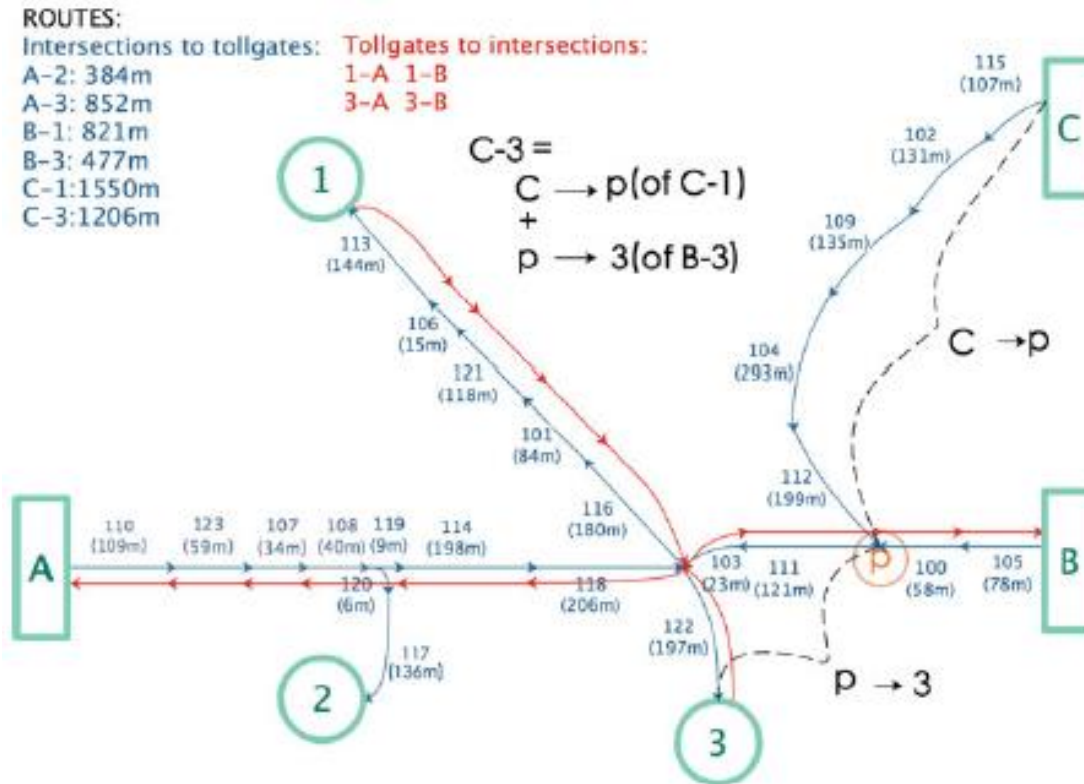
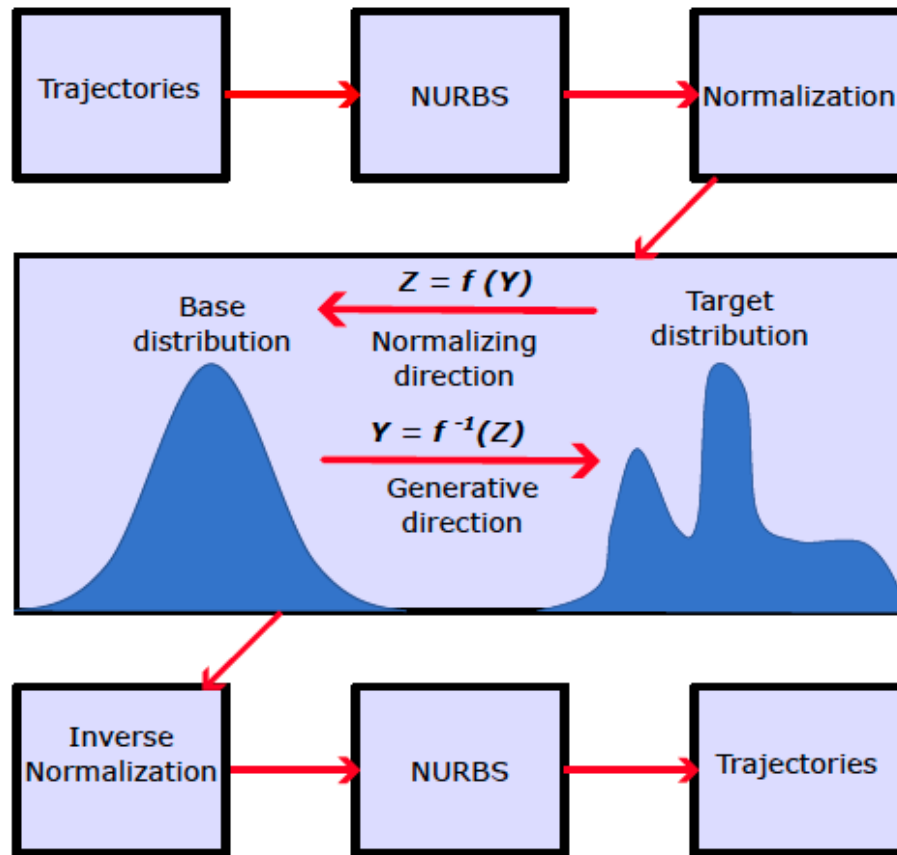


Fig. 1 The link representation of road network. Each route is composed by a sequence of links and each link is represented by an arrow. The value without parentheses over a link represents the unique ID of the link and the value in parentheses represents the length of the link

- Travel time prediction
- Impute missing data using domain knowledge

Synthetic data



- Adapt generative models to generate synthetic data (e.g., vehicle trajectories) -> sample the rare cases more frequently
- Demonstrated the method on cut-in scenario

Hyperparameters tuning

Hyperparameters vs parameters

Hyperparameters

- Often used to help estimate the model parameters
- Specified by the users
- Tuned for the problem at hand
- In random forests: number of trees, number of features considered
- In NN: learning rate, number of hidden layers

Parameters

- Are needed by the model for making predictions
- The values are estimated from the data
- Usually are not set by the users
- In NN: weights

Tuning the hyperparameters

Methods for tuning hyperparameters

- Trial and error
- Grid search (e.g., [GridSearchCV](#))
- Randomised search (e.g., [RandomizedSearchCV](#))
- Via optimisation (e.g., [BayesSeachCV](#))

Algorithms for tuning hyperparameters:

- [population-based training](#) (for NN)
- [Sequential model-based optimisation](#)

More information: [here](#) and [here](#).

See notebook!

Next class session

- Preprocessing data
- Feature extraction
- Feature selection